

Mathematics Foundations - Class 8

Original copyright-free textbook for AI-teacher evaluation.

Chapter 1: Rational Numbers

Chapter 1: Rational Numbers Section 1: Definition of rational numbers, properties, number line representation, equivalent forms. Addition, subtraction, multiplication and division with step-by-step examples. Applications in daily life. Detailed explanation: Students should understand concepts, observe patterns, solve guided examples, and connect mathematics with real-life situations. Worked Example: Demonstrate each step clearly and justify the operation used.

Chapter 1: Rational Numbers Section 2: Definition of rational numbers, properties, number line representation, equivalent forms. Addition, subtraction, multiplication and division with step-by-step examples. Applications in daily life. Detailed explanation: Students should understand concepts, observe patterns, solve guided examples, and connect mathematics with real-life situations. Worked Example: Demonstrate each step clearly and justify the operation used.

Chapter 1: Rational Numbers Section 3: Definition of rational numbers, properties, number line representation, equivalent forms. Addition, subtraction, multiplication and division with step-by-step examples. Applications in daily life. Detailed explanation: Students should understand concepts, observe patterns, solve guided examples, and connect mathematics with real-life situations. Worked Example: Demonstrate each step clearly and justify the operation used.

Chapter 1: Rational Numbers Section 4: Definition of rational numbers, properties, number line representation, equivalent forms. Addition, subtraction, multiplication and division with step-by-step examples. Applications in daily life. Detailed explanation: Students should understand concepts, observe patterns, solve guided examples, and connect mathematics with real-life situations. Worked Example: Demonstrate each step clearly and justify the operation used.

Solved Examples

Example: Find $\frac{3}{4} + \frac{5}{8}$

Solution: Show each mathematical step logically.

Example: Simplify $-\frac{2}{3} \times \frac{9}{4}$

Solution: Show each mathematical step logically.

Exercises

1. Practice question for Chapter 1: Rational Numbers.
2. Practice question for Chapter 1: Rational Numbers.
3. Practice question for Chapter 1: Rational Numbers.
4. Practice question for Chapter 1: Rational Numbers.
5. Practice question for Chapter 1: Rational Numbers.
6. Practice question for Chapter 1: Rational Numbers.
7. Practice question for Chapter 1: Rational Numbers.
8. Practice question for Chapter 1: Rational Numbers.
9. Practice question for Chapter 1: Rational Numbers.
10. Practice question for Chapter 1: Rational Numbers.

HOTS Questions

HOTS 1: Multi-step reasoning problem related to Chapter 1: Rational Numbers.

HOTS 2: Multi-step reasoning problem related to Chapter 1: Rational Numbers.

HOTS 3: Multi-step reasoning problem related to Chapter 1: Rational Numbers.

Chapter 2: Linear Equations in One Variable

Chapter 2: Linear Equations in One Variable Section 1: Understanding variables and constants. Balancing method. Word problems. Detailed explanation: Students should understand concepts, observe patterns, solve guided examples, and connect mathematics with real-life situations. Worked Example: Demonstrate each step clearly and justify the operation used.

Chapter 2: Linear Equations in One Variable Section 2: Understanding variables and constants. Balancing method. Word problems. Detailed explanation: Students should understand concepts, observe patterns, solve guided examples, and connect mathematics with real-life situations. Worked Example: Demonstrate each step clearly and justify the operation used.

Chapter 2: Linear Equations in One Variable Section 3: Understanding variables and constants. Balancing method. Word problems. Detailed explanation: Students should understand concepts, observe patterns, solve guided examples, and connect mathematics with real-life situations. Worked Example: Demonstrate each step clearly and justify the operation used.

Chapter 2: Linear Equations in One Variable Section 4: Understanding variables and constants. Balancing method. Word problems. Detailed explanation: Students should understand concepts, observe patterns, solve guided examples, and connect mathematics with real-life situations. Worked Example: Demonstrate each step clearly and justify the operation used.

Solved Examples

Example: Solve $3x+5=20$

Solution: Show each mathematical step logically.

Example: A number increased by 7 becomes 25. Find the number.

Solution: Show each mathematical step logically.

Exercises

1. Practice question for Chapter 2: Linear Equations in One Variable.
2. Practice question for Chapter 2: Linear Equations in One Variable.
3. Practice question for Chapter 2: Linear Equations in One Variable.
4. Practice question for Chapter 2: Linear Equations in One Variable.
5. Practice question for Chapter 2: Linear Equations in One Variable.
6. Practice question for Chapter 2: Linear Equations in One Variable.
7. Practice question for Chapter 2: Linear Equations in One Variable.
8. Practice question for Chapter 2: Linear Equations in One Variable.
9. Practice question for Chapter 2: Linear Equations in One Variable.
10. Practice question for Chapter 2: Linear Equations in One Variable.

HOTS Questions

HOTS 1: Multi-step reasoning problem related to Chapter 2: Linear Equations in One Variable.

HOTS 2: Multi-step reasoning problem related to Chapter 2: Linear Equations in One Variable.

HOTS 3: Multi-step reasoning problem related to Chapter 2: Linear Equations in One Variable.

Chapter 3: Algebraic Expressions and Identities

Chapter 3: Algebraic Expressions and Identities Section 1: Terms, coefficients, like and unlike terms. Expansion and simplification. Detailed explanation: Students should

understand concepts, observe patterns, solve guided examples, and connect mathematics with real-life situations. Worked Example: Demonstrate each step clearly and justify the operation used.

Chapter 3: Algebraic Expressions and Identities Section 2: Terms, coefficients, like and unlike terms. Expansion and simplification. Detailed explanation: Students should understand concepts, observe patterns, solve guided examples, and connect mathematics with real-life situations. Worked Example: Demonstrate each step clearly and justify the operation used.

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Chapter 3: Algebraic Expressions and Identities Section 4: Terms, coefficients, like and unlike terms. Expansion and simplification. Detailed explanation: Students should understand concepts, observe patterns, solve guided examples, and connect mathematics with real-life situations. Worked Example: Demonstrate each step clearly and justify the operation used.

Solved Examples

Example: Simplify $2a+5a-3a$

Solution: Show each mathematical step logically.

Example: Expand $(x+2)(x+3)$

Solution: Show each mathematical step logically.

Exercises

1. Practice question for Chapter 3: Algebraic Expressions and Identities.
2. Practice question for Chapter 3: Algebraic Expressions and Identities.
3. Practice question for Chapter 3: Algebraic Expressions and Identities.
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8. Practice question for Chapter 3: Algebraic Expressions and Identities.
9. Practice question for Chapter 3: Algebraic Expressions and Identities.

10. Practice question for Chapter 3: Algebraic Expressions and Identities.

HOTS Questions

HOTS 1: Multi-step reasoning problem related to Chapter 3: Algebraic Expressions and Identities.

HOTS 2: Multi-step reasoning problem related to Chapter 3: Algebraic Expressions and Identities.

HOTS 3: Multi-step reasoning problem related to Chapter 3: Algebraic Expressions and Identities.

Chapter 4: Geometry

Chapter 4: Geometry Section 1: Lines, angles, polygons, triangles, quadrilaterals. Angle properties and reasoning. Detailed explanation: Students should understand concepts, observe patterns, solve guided examples, and connect mathematics with real-life situations. Worked Example: Demonstrate each step clearly and justify the operation used.

Chapter 4: Geometry Section 2: Lines, angles, polygons, triangles, quadrilaterals. Angle properties and reasoning. Detailed explanation: Students should understand concepts, observe patterns, solve guided examples, and connect mathematics with real-life situations. Worked Example: Demonstrate each step clearly and justify the operation used.

Chapter 4: Geometry Section 3: Lines, angles, polygons, triangles, quadrilaterals. Angle properties and reasoning. Detailed explanation: Students should understand concepts, observe patterns, solve guided examples, and connect mathematics with real-life situations. Worked Example: Demonstrate each step clearly and justify the operation used.

Chapter 4: Geometry Section 4: Lines, angles, polygons, triangles, quadrilaterals. Angle properties and reasoning. Detailed explanation: Students should understand concepts, observe patterns, solve guided examples, and connect mathematics with real-life situations. Worked Example: Demonstrate each step clearly and justify the operation used.

Solved Examples

Example: Find missing angle in a triangle.

Solution: Show each mathematical step logically.

Exercises

1. Practice question for Chapter 4: Geometry.
2. Practice question for Chapter 4: Geometry.
3. Practice question for Chapter 4: Geometry.
4. Practice question for Chapter 4: Geometry.

5. Practice question for Chapter 4: Geometry.
6. Practice question for Chapter 4: Geometry.
7. Practice question for Chapter 4: Geometry.
8. Practice question for Chapter 4: Geometry.
9. Practice question for Chapter 4: Geometry.
10. Practice question for Chapter 4: Geometry.

HOTS Questions

HOTS 1: Multi-step reasoning problem related to Chapter 4: Geometry.

HOTS 2: Multi-step reasoning problem related to Chapter 4: Geometry.

HOTS 3: Multi-step reasoning problem related to Chapter 4: Geometry.

Chapter 5: Mensuration

Chapter 5: Mensuration Section 1: Perimeter, area, volume. Real-world measurement problems. Detailed explanation: Students should understand concepts, observe patterns, solve guided examples, and connect mathematics with real-life situations. Worked Example: Demonstrate each step clearly and justify the operation used.

Chapter 5: Mensuration Section 2: Perimeter, area, volume. Real-world measurement problems. Detailed explanation: Students should understand concepts, observe patterns, solve guided examples, and connect mathematics with real-life situations. Worked Example: Demonstrate each step clearly and justify the operation used.

Chapter 5: Mensuration Section 3: Perimeter, area, volume. Real-world measurement problems. Detailed explanation: Students should understand concepts, observe patterns, solve guided examples, and connect mathematics with real-life situations. Worked Example: Demonstrate each step clearly and justify the operation used.

Chapter 5: Mensuration Section 4: Perimeter, area, volume. Real-world measurement problems. Detailed explanation: Students should understand concepts, observe patterns, solve guided examples, and connect mathematics with real-life situations. Worked Example: Demonstrate each step clearly and justify the operation used.

Solved Examples

Example: Area of rectangle $12\text{ cm} \times 8\text{ cm}$

Solution: Show each mathematical step logically.

Exercises

1. Practice question for Chapter 5: Mensuration.

2. Practice question for Chapter 5: Mensuration.
3. Practice question for Chapter 5: Mensuration.
4. Practice question for Chapter 5: Mensuration.
5. Practice question for Chapter 5: Mensuration.
6. Practice question for Chapter 5: Mensuration.
7. Practice question for Chapter 5: Mensuration.
8. Practice question for Chapter 5: Mensuration.
9. Practice question for Chapter 5: Mensuration.
10. Practice question for Chapter 5: Mensuration.

HOTS Questions

HOTS 1: Multi-step reasoning problem related to Chapter 5: Mensuration.

HOTS 2: Multi-step reasoning problem related to Chapter 5: Mensuration.

HOTS 3: Multi-step reasoning problem related to Chapter 5: Mensuration.

Chapter 6: Data Handling and Probability

Chapter 6: Data Handling and Probability Section 1: Tables, bar graphs, averages, probability basics. Detailed explanation: Students should understand concepts, observe patterns, solve guided examples, and connect mathematics with real-life situations. Worked Example: Demonstrate each step clearly and justify the operation used.

Chapter 6: Data Handling and Probability Section 2: Tables, bar graphs, averages, probability basics. Detailed explanation: Students should understand concepts, observe patterns, solve guided examples, and connect mathematics with real-life situations. Worked Example: Demonstrate each step clearly and justify the operation used.

Chapter 6: Data Handling and Probability Section 3: Tables, bar graphs, averages, probability basics. Detailed explanation: Students should understand concepts, observe patterns, solve guided examples, and connect mathematics with real-life situations. Worked Example: Demonstrate each step clearly and justify the operation used.

Chapter 6: Data Handling and Probability Section 4: Tables, bar graphs, averages, probability basics. Detailed explanation: Students should understand concepts, observe patterns, solve guided examples, and connect mathematics with real-life situations. Worked Example: Demonstrate each step clearly and justify the operation used.

Solved Examples

Example: Find mean of 10,12,15,18,20

Solution: Show each mathematical step logically.

Exercises

1. Practice question for Chapter 6: Data Handling and Probability.
2. Practice question for Chapter 6: Data Handling and Probability.
3. Practice question for Chapter 6: Data Handling and Probability.
4. Practice question for Chapter 6: Data Handling and Probability.
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8. Practice question for Chapter 6: Data Handling and Probability.
9. Practice question for Chapter 6: Data Handling and Probability.
10. Practice question for Chapter 6: Data Handling and Probability.

HOTS Questions

HOTS 1: Multi-step reasoning problem related to Chapter 6: Data Handling and Probability.

HOTS 2: Multi-step reasoning problem related to Chapter 6: Data Handling and Probability.

HOTS 3: Multi-step reasoning problem related to Chapter 6: Data Handling and Probability.

Answer Key

Answer reference item 1: Sample solution approach.

Answer reference item 2: Sample solution approach.

Answer reference item 3: Sample solution approach.

Answer reference item 4: Sample solution approach.

Answer reference item 5: Sample solution approach.

Answer reference item 6: Sample solution approach.

Answer reference item 7: Sample solution approach.

Answer reference item 8: Sample solution approach.

Answer reference item 9: Sample solution approach.

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Answer reference item 54: Sample solution approach.

Answer reference item 55: Sample solution approach.

Answer reference item 56: Sample solution approach.

Answer reference item 57: Sample solution approach.

Answer reference item 58: Sample solution approach.

Answer reference item 59: Sample solution approach.

Answer reference item 60: Sample solution approach.